

Fortessa Standard Operating Procedure & User Guide – Plate Reader
UC Davis Flow Cytometry Shared Resource

Updated 01/12/2021

1. Complete both Fortessa Startup and DIVA SOPs before continuing.
2. Place the Fortessa on **STANDBY**
3. Switch from tube to plate-based acquisition (toggle switch on left side of instrument)
4. Remove water tube from SIP – leave the arm to the side
5. Unscrew droplet containment SIP sleeve. (Small plastic cone shaped object at the top of the SIT).
6. Install and screw on HTS SIT sleeve (shorter version of the SIP sleeve for tubes, found in drawer on right hand side) – finger tight, do NOT over tighten.
7. Locate the HTS sample line (peak tubing coming from HTS with SIT adaptor on the end)
8. Loosen the nut at the top of the HTS SIT adaptor
9. Install the HTS SIT adaptor by slipping the SIT into the adaptor.
10. While pressing up lightly (to ensure adaptor stays fully seated against bottom of SIT) finger tighten the adaptor nut.
11. Place the Fortessa on **LOW**
12. Turn plate reader on. ***Note: Speed of sample acquisition on HTS is determined by injection rate (set in software) not flow rate setting on Fortessa control panel like tube-based acquisition.*
13. Place plate on plate reader. Be sure that well A1 is aligned appropriately and filled with 210ul of water.
14. Place the Fortessa into **RUN**
15. Prime the HTS prior to first use.
 - a. In DIVA at the top select HTS -> Prime
 - b. Repeat this priming step until all air is out of HTS fluid lines.
16. Place the Fortessa into **STANDBY**
17. Enable the plate window. 
18. Create a new plate in your experiment
 - a. Select the plate type you are running
 - b. Press the new plate button
19. General settings for plate reader should be:
 - a. Throughput Mode: Standard (not High which is default)
 - b. Sample Flow Rate: See below
 - i. The lowest setting will yield the best resolution for your data. Higher flow rates can be used if population signal separation allows –see b and c. Try not to run over 8000 events/s to avoid clogging
 - ii. Based on cell size, sample density, and “stickiness”. In general upper limits for splenocytes can be run at 2.0-2.5ul/s, tumor cells must be run on a slower flow rate of 1.5ul/s, and large granular cells such as hepatocytes must be run on the lowest speed as these will have a large propensity to clog the machine.

- c. Sample Volume: Varies, **be sure that it's at least 50ul less** than total well volume)
 - d. Mix Volume: 50-80ul (should be 100 less than your sample volume)
 - e. Mix Speed: 80ul/s (maximum)
 - f. # of Mixes: 5 (maximum)
 - g. Wash Volume: 700ul (maximum)
20. Highlight wells that you would like to run on plate window and set up your plate.
 21. Go to experiment layout to enter key information such as antibody panels, key words, and number of cells to collect per well. If you do not do this, the default is only 10,000 cells per well.
 22. Place the Fortessa into **RUN**
 23. Next highlight all your wells again, ensure that they have the correct run parameters. Select **"Run Wells"** from acquisition dashboard. This will begin your plate run.
 - a. **DO NOT** select **"Run Plate"**, this will also run any empty wells that you have in your plate, thereby introducing air into the system.
 24. Once the plate finishes, place the Fortessa into **STANDBY**.
 25. Remove cover to plate reader, remove your finished plate.
 26. If running **multiple plates**:
 - a. You **MUST** run an abridged cleaning plate between each plate
 - i. **Create abridged cleaning plate:** Into a cleaning plate fill 5 wells with 10% bleach and 5 wells with DI water
 - ii. If not already present create a new plate in software (step 18) with the following settings
 - a. Highest flow rate(3uL/s)
 - b. Throughput Mode: Standard (not High which is default)
 - c. Sample Flow Rate: 3.0ul/s
 - d. Sample Volume: 150ul
 - e. Mix Volume: 80ul
 - f. Mix Speed: 80ul/s (maximum)
 - g. # of Mixes: 5 (maximum)
 - h. Wash Volume: 700ul (maximum)
 - b. Place the Fortessa into **RUN**
 - c. Highlight the cleaning wells that you would like to run on plate window.
 - d. Select **"Run Wells"** from acquisition dashboard. This will begin your cleaning plate run.
 - e. When the cleaning is finished Place the Fortessa into **STANDBY**
 - f. **REFILL** the sheath tank to the weld line ***Note: Sheath will be pressurized. You must vent before opening.*
 - i. This needs to be done after **each** plate!
 27. Repeat from step 18-26 until all plates are run. ***Be sure to run the abridged cleaning plate and refill sheath between plates to ensure it doesn't run dry!!***
 28. Once the final plate is finished, place Fortessa into **STANDBY**.

Fortessa HTS Cleaning:

29. **Create FULL cleaning plate:** Into a new cleaning plate fill 4 wells with FACS Rinse, 4 wells with 10% bleach, 4 wells with FACS Rinse, and 8 wells with DI water.

30. Plate cleaning plate onto HTS.
31. Create new plate within DIVA (step 18) with 4 wells FACSRinse, 4 wells 10% bleach, 4 wells FACSRinse, and 8 wells of water.
32. Run the clean plate at highest flow rate(3uL/s) using following settings:
 - a. Throughput Mode: Standard (not High which is default)
 - b. Sample Flow Rate: 3.0ul/s
 - c. Sample Volume: 150ul
 - d. Mix Volume: 80ul
 - e. Mix Speed: 80ul/s (maximum)
 - f. # of Mixes: 5 (maximum)
 - g. Wash Volume: 700ul (maximum)
33. Place the Fortessa into **RUN**
34. Highlight the cleaning wells that you would like to run on plate window.
35. Select "**Run Wells**" from acquisition dashboard. This will begin your cleaning plate run.
36. Once the cleaning plate is finished, place Fortessa into **STANDBY**.
37. Unplug plate from HTS sleeve. Turn HTS off and switch Fortessa back to tube-based acquisition.
38. Replace HTS sleeve with normal SIP sleeve.
39. After finishing the plate wash cycles, you **MUST** run the tube-based cleaning:
 - a. Prepare 3 new tubes: 1 each of 10% bleach, FACSRinse, and Water.
 - b. Place FACSRinse tube onto SIP. Switch Fortessa to HIGH speed and run.
 - i. To visualize events: Create a new specimen entitled Clean, select the first tube and hit acquire.
 - ii. Run FACSRinse x 5minutes.
 - c. Switch to 10% bleach. Run x5 minutes.
 - d. Switch to water. Run x5 minutes. Events should be ~>30 events/s.
 - e. Switch Fortessa back to low speed and standby.
 - f. Empty waste and fill sheath tank to weld line.
 - g. Power off Fortessa if no users coming within 2hr.